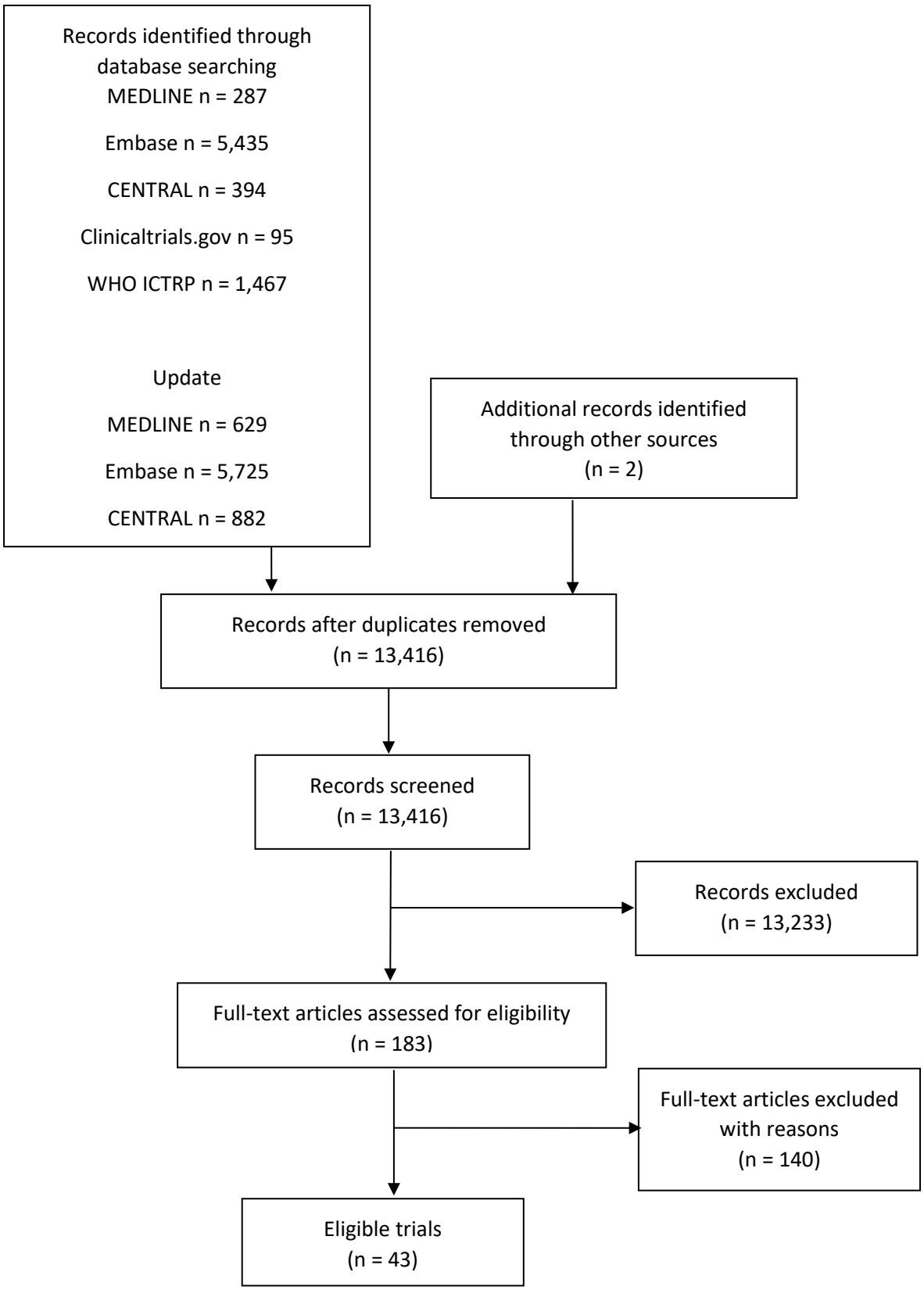
Supplement 4. eResults

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1. Flow diagram



2. Characteristics of included studies

Where age or severity score was reported for each group, the reported values were combined. Where a mean value with standard deviation was unavailable, but a median with interquartile range was available, the mean and SD were calculated using formulae[1,2]. If neither of the measures mentioned above was available, the mean value of known SDs was calculated from included studies[3]. GC denotes glucocorticoid; vitC, vitamin C; VHD-vitC, very high dose vitamin C; HD-vitC, high dose vitamin C; vitB1, vitamin B1; N/A, not available; SAPS, Simplified Acute Physiology Score; APACHE, Acute Physiology And Chronic Health Evaluation.

Study Author year	Age, mean (SD)	Shock	Severity score, mean	Industry-funded	Country	Comparison
Bennett Jr 1963[4]	N/A	Yes	N/A	Unclear	US	GC vs placebo or usual care
Bollaert 1998[5]	58 (14.6)	Yes	SAPS 2, 14	No	France	GC vs placebo or usual care
Annane 2002[6]	61 (16)	Yes	SAPS 2, 58.5	No	France	GC vs placebo or usual care
Yildiz 2002[7]	57.1 (16.9)	No	APACHE II, 16.6	Unclear	Turkey	GC vs placebo or usual care
Oppert 2005[8]	N/A	Yes	N/A	Unclear	Germany	GC vs placebo or usual care
Rinaldi 2006[9]	N/A	No	N/A	Unclear	Italy	GC vs placebo or usual care
Kaufmann 2008[10]	62.3 (17.9)	Yes	APACHE III, 55.2	Unclear	Germany	GC vs placebo or usual care
Sprung 2008[11] **(CORTICUS)**	63 (14.5)	Yes	SAPS 2, 49.1	Yes	France, Germany, et al.	GC vs placebo or usual care
Ferron-Celma 2009[12]	66.5 (4.2)	No	N/A	No	Spain	vitC vs placebo or usual care
Hu 2009[13]	55.3 (36.6)	Yes	APACHE II, 19.1	Unclear	China	GC vs placebo or usual care
Meduri 2009[14]	N/A	No	APACHE III, 71.4	Unclear	US	GC vs placebo or usual care
Arabi 2010[15]	60 (12.3)	Yes	APACHE II, 29.7	No	Saudi Arabia	GC vs placebo or usual care
Deng 2011[16]	N/A	No	N/A	Unclear	China	GC vs placebo or usual care
Yildiz 2011[17]	62.8 (15.6)	No	APACHE II, 20.7	Unclear	Turkey	GC vs placebo or usual care
Rahardjo 2013[18]	N/A	No	N/A	Unclear	Indonesia	vitC vs placebo or usual care
Fowler 2014[19]	N/A	No	N/A	No	US	VHD-vitC vs vitC vs placebo or usual care
Gordon 2014[20]	61.2 (17)	Yes	N/A	No	UK	GC vs placebo or usual care
Mirea 2014[21]	N/A	No	N/A	Unclear	Romania	GC (intermittent vs continuous) vs placebo or usual care
Donnino 2016[22]	67.4 (15.7)	Yes	APACHE II, 26.1	No	US	vitB1 vs placebo or usual care
Gordon 2016[23] **(VANISH)**	65.6 (17.1)	Yes	APACHE II, 24.4	No	UK	GC vs placebo or usual care
Keh 2016[24] **(HYPRESS)**	65 (14.4)	No	APACHE II, 19	No	Germany	GC vs placebo or usual care
Li 2016[25]	66.1 (10.1)	No	N/A	No	China	GC vs placebo or usual care

Tongyoo 2016[26]	64.4 (16.6)	No	APACHE II, 21.8	No	Thailand	GC vs placebo or usual care
Zabet 2016[27]	63.9 (14.7)	Yes	APACHE II, 21	No	Iran	HD-vitC vs placebo or usual care
Lv 2017[28]	66.8 (14.9)	No	APACHE II, 23.4	Unclear	China	GC vs placebo or usual care
Annane 2018[29] **(APROCCHSS)**	66 (14)	Yes	SAPS 2, 56	No	France	GC vs placebo or usual care
Balakrishnan 2018[30]	54.4 (12.3)	Yes	APACHE III, 56.4	No	India	HD-vitC + GC + vitB1 vs placebo or usual care
Rosini 2018[31]	N/A	No	N/A	No	US	VHD-vitC vs placebo or usual care
Venkatesh 2018[32] **(ADRENAL)**	62.5 (15)	Yes	N/A	Yes	Australia, UK, et al.	GC vs placebo or usual care
Fowler 2019[33] **(CITRIS-ALI)**	55.1 (20.4)	No	N/A	Yes	USA	VHD-vitC vs placebo or usual care
Jiang 2019[34]	59.2 (10.3)	Yes	APACHE II, 30.5	No	China	GC (early vs late) vs placebo or usual care
Harun 2019[35]	63.7 (17.7)	Yes	N/A	No	Malaysia	vitB1 vs placebo or usual care
Chang 2020[36] **(HYVCTTSSS)**	61.6 (14)	No	APACHE II, 23	Unclear	China	HD-vitC + GC + vitB1 vs placebo or usual care
Fujii & Luethi 2020[37] **(VITAMINS)**	61.7 (15)	Yes	APACHE III, 80.3	No	Australia, New Zealand, et al.	HD-vitC + GC + vitB1 vs GC
Hwang 2020[38] **(ATESS)**	48.4 (9.9)	Yes	N/A	No	South Korea	HD-vitC + vitB1 vs placebo or usual care
Iglesias 2020[39] **(ORANGES)**	69 (13)	No	APACHE II, 24.5	No	US	HD-vitC + GC + vitB1 vs placebo or usual care
Lv 2020[40]	59.4 (14.2)	No	N/A	Unclear	China	HD-vitC vs placebo or usual care
Mohamed 2020[41] **(ViCTOR)**	59 (14.9)	Yes	N/A	No	India	HD-vitC + GC + vitB1 vs placebo or usual care
Moskowitz 2020[42] **(ACTS)**	68.3 (14.4)	Yes	N/A	No	US	HD-vitC + GC + vitB1 vs placebo or usual care
Petsakul 2020[43]	65 (17.8)	Yes	APACHE II, 27.5	No	Thailand	vitB1 vs placebo or usual care
Wani 2020[44]	54.1 (35.7)	No	N/A	No	India	HD-vitC + GC + vitB1 vs placebo or usual care
Sevransky 2021[45] **(VICTAS)**	60.8 (15)	No	APACHE II, 26.9	Yes	US	HD-vitC + GC + vitB1 vs placebo or usual care
Hussain 2021[46]	63.9 (16.3)	Yes	APACHE II, 20.6	Unclear	Egypt	HD-vitC + GC + vitB1 vs GC

2.1 Comments in assessing Risk of Bias

Study	Comments
Bennett Jr 1963[4]	year 1963 study, No protocol, About 15 patients < 18 years old *(Selective reporting: Unclear, General comment)*
Rinaldi 2006[9]	Too many dropouts. *(Incomplete outcome data on the all-cause mortality within 1 year but later than 90 days: Unclear, Fraud or concern that might be a source of risk of bias, other bias)*
Sprung 2008[11] (CORTICUS)	Based on the number of missingness and outcomes, incomplete outcome or 21 open label HC infusion are considered to be at low risk of bias. *(Incomplete outcome data on the all-cause mortality within 1 year but later than 90 days: Low)* Early termination, but without positive results. *(Fraud or concern that might be a source of risk of bias, other bias)*
Arabi 2010[15]	Study interrupted for futility after enrolment of 50% sample size *(Fraud or concern that might be a source of risk of bias, reported in free text, other bias)*
Balakrishnan 2018[30]	Very poor information on baseline characteristics and poor outcome reporting *(Selective reporting: High, Baseline imbalance: Unclear)*
Harun 2019[35]	protocol or trial registry is not available. *(Selective reporting: Unclear)* Baseline APAHCE2 was higher in the Thiamine group *(Baseline imbalance: High)* Lack of ITT analysis. 7 patients died after randomization and were excluded. *(General comment)*
Chang 2020[36] (HYVCTTSSS)	Placebo was not administered in 70% patients in the control group *(Inappropriate study deviation: High)*
Iglesias 2020[39] (ORANGES)	https://clinicaltrials.gov/ct2/show/NCT03422159 See history. *(Selective reporting: High)* Baseline: patients requiring vasopressors were unbalanced *(Baseline imbalance: High)*
Wani 2020[44]	requirement for renal replacement therapy, which was planned to measure in the protocol, was not reported in the publication. *(Selective reporting: High)*
Sevransky 2021[45] (VICTAS)	Sponsor: Nova and some companies seem to be involved? (providing drugs) *(Industrial sponsorship)* Deviation: the number of patients who did not receive assigned treatment was not balanced in the 2 groups. But I could not tell whether it was by chance or not and its consequence. If it was perfectly blinded, I would say it was may be due to chance and may not be cause big bias. Thirty-three percent of patients in the intervention group and 32% of control patients received clinician-prescribed corticosteroids at a dose of at least 200 mg of hydrocortisone daily equivalent. If corticosteroids have a beneficial effect on VVFD, this would bias the trial results toward the null, although a post hoc sensitivity analysis demonstrated similar findings in patients not receiving open-label corticosteroids. *(Inappropriate study deviation: Unclear)*
Hussain 2021[46]	significant drop-out *(Incomplete outcome data: Unclear)* Lack ITT analysis *(General comment)*

3. Assessment of transitivity: distribution of effect modifiers

We assessed whether the eligible trials were similar in terms of characteristics that might modify the treatment effect on average so that the transitivity assumption was plausible. The following characteristics were assessed as possible effect modifiers: industrial sponsorship, blinding of the personnel, and vasopressor dependency of the study population. The plausibility of the transitivity assumption was evaluated by comparing the distribution of the effect modifiers across studies grouped by intervention arms through visual inspection. GC denotes glucocorticoid; vitC, vitamin C; VHD-vitC, very high dose vitamin C; HD-vitC, high dose vitamin C; vitB1, vitamin B1.

Comparison	Study	Industry-funded	Shock	Study personnel blinded
HD-vitC + GC + vitB1 vs GC	Fujii & Luethi 2020	No or Unclear	Yes	No or Unclear
HD-vitC + GC + vitB1 vs GC	Hussein 2021	No or Unclear	Yes	No or Unclear
HD-vitC + GC + vitB1 vs placebo or usual care	Balakrishnan 2018	No or Unclear	Yes	Yes
HD-vitC + GC + vitB1 vs placebo or usual care	Chang 2020	No or Unclear	No	No or Unclear
HD-vitC + GC + vitB1 vs placebo or usual care	Iglesias 2020	No or Unclear	No	Yes
HD-vitC + GC + vitB1 vs placebo or usual care	Mohamed 2020	No or Unclear	Yes	No or Unclear
HD-vitC + GC + vitB1 vs placebo or usual care	Moskowitz 2020	No or Unclear	Yes	Yes
HD-vitC + GC + vitB1 vs placebo or usual care	Wani 2020	No or Unclear	No	No or Unclear
HD-vitC + GC + vitB1 vs placebo or usual care	Sevransky 2021	No or Unclear	No	Yes
VHD-vitC vs vitC vs placebo or usual care	Fowler 2014	No or Unclear	No	Yes
HD-vitC + vitB1 vs placebo or usual care	Hwang 2020	No or Unclear	Yes	Yes
VHD-vitC vs placebo or usual care	Rosini 2018	No or Unclear	No	Yes
VHD-vitC vs placebo or usual care	Fowler 2019	Yes	No	Yes
HD-vitC vs placebo or usual care	Zabet 2016	No or Unclear	Yes	Yes
HD-vitC vs placebo or usual care	Lv 2020	No or Unclear	No	No or Unclear
vitC vs placebo or usual care	Ferron-Celma 2009	No or Unclear	No	Yes
vitC vs placebo or usual care	Rahardjo 2013	No or Unclear	No	Yes
GC vs placebo or usual care	Bennett Jr 1963	No or Unclear	Yes	Yes
GC vs placebo or usual care	Bollaert 1998	No or Unclear	Yes	Yes
GC vs placebo or usual care	Annane 2002	No or Unclear	Yes	Yes
GC vs placebo or usual care	Yildiz 2002	No or Unclear	No	Yes
GC vs placebo or usual care	Oppert 2005	No or Unclear	Yes	Yes
GC vs placebo or usual care	Rinaldi 2006	No or Unclear	No	No or Unclear
GC vs placebo or usual care	Kaufmann 2008	No or Unclear	Yes	Yes
GC vs placebo or usual care	Sprung 2008	Yes	Yes	Yes
GC vs placebo or usual care	Hu 2009	No or Unclear	Yes	No or Unclear
GC vs placebo or usual care	Meduri 2009	No or Unclear	No	Yes
GC vs placebo or usual care	Arabi 2010	No or Unclear	Yes	Yes

GC vs placebo or usual care	Deng 2011	No or Unclear	No	No or Unclear
GC vs placebo or usual care	Yildiz 2011	No or Unclear	No	Yes
GC vs placebo or usual care	Gordon 2014	No or Unclear	Yes	No or Unclear
GC vs placebo or usual care	Mirea 2014	No or Unclear	No	No or Unclear
GC vs placebo or usual care	Gordon 2016	No or Unclear	Yes	Yes
GC vs placebo or usual care	Keh 2016	No or Unclear	No	Yes
GC vs placebo or usual care	Li 2016	No or Unclear	No	No or Unclear
GC vs placebo or usual care	Tongyoo 2016	No or Unclear	No	Yes
GC vs placebo or usual care	Lv 2017	No or Unclear	No	Yes
GC vs placebo or usual care	Annane 2018	No or Unclear	Yes	Yes
GC vs placebo or usual care	Venkatesh 2018	Yes	Yes	Yes
GC vs placebo or usual care	Jiang 2019	No or Unclear	Yes	No or Unclear
vitB1 vs placebo or usual care	Donnino 2016	No or Unclear	Yes	Yes
vitB1 vs placebo or usual care	Petsakul 2020	No or Unclear	Yes	Yes
vitB1 vs placebo or usual care	Harun 2019	No or Unclear	Yes	No or Unclear

4. All-cause mortality from 90 days to 1 year

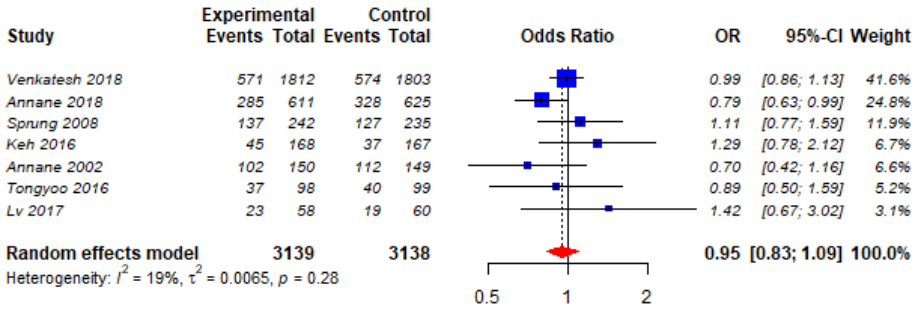
4.1. Heterogeneity: Between-study variance within a comparison and the network

OR denotes odds ratio; GC, glucocorticoid; HD-vitC, high dose vitamin C; vitB1, vitamin B1; N/A, not available.

Comparisons	Number of studies	Expected variance for ORs from empirical distribution[47] median [95% range]	Estimated variance		Forest plot
			in comparison	in network	
HD-vitC+GC+vitB1 vs GC	1	0.014 [0.0008 to 0.25]	N/A	0.0018	N/A
HD-vitC+GC+vitB1 vs placebo or usual care	1	0.017 [0.001 to 0.30]	N/A	0.0018	N/A
HD-vitC+vitB1 vs placebo or usual care	1	0.017 [0.001 to 0.30]	N/A	0.0018	N/A
GC vs placebo or usual care	7	0.017 [0.001 to 0.30]	0.0065	0.0018	No heterogeneity

Forest plot of pairwise comparison

GC vs placebo or usual care



4.2. Inconsistency

4.2.1. side-split

Number of loops = 1. OR denotes odds ratio; RoR, Ratio of Ratios (direct versus indirect); GC, glucocorticoid; HD-vitC, high dose vitamin C; vitB1, vitamin B1.

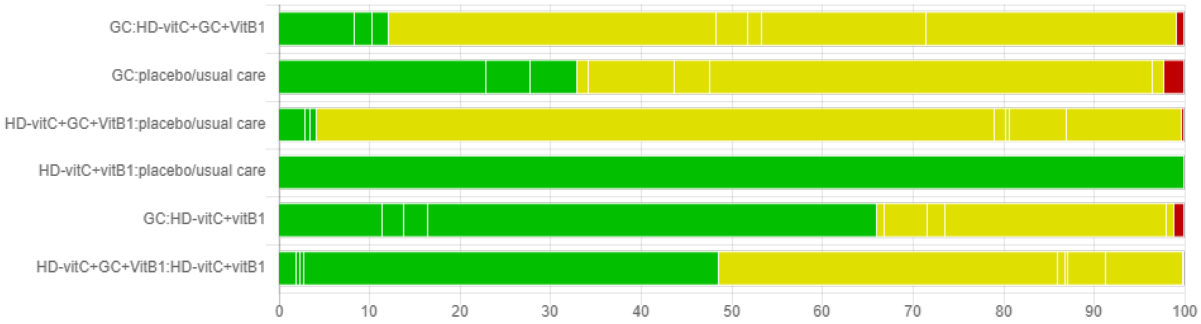
comparison	Number of studies	OR from NMA	OR from direct comparison	OR from indirect comparison	RoR	p-value
HD-vitC+GC+vitB1:GC	1	1.2	1.23	1.18	1.04	0.9138
HD-vitC+GC+vitB1:placebo or usual care	1	1.13	1.12	1.17	0.96	0.9138
HD-vitC+vitB1:GC	0	1.31	.	1.31	.	.
HD-vitC+vitB1:HD-vitC+GC+vitB1	0	1.09	.	1.09	.	.
HD-vitC+vitB1:placebo or usual care	1	1.24	1.24	.	.	.
GC:placebo or usual care	7	0.95	0.95	0.91	1.04	0.9138

4.2.2. global

Q statistic to assess consistency under the assumption of a full design-by-treatment interaction random effects model = 0.01 (df=1, p=0.9159)

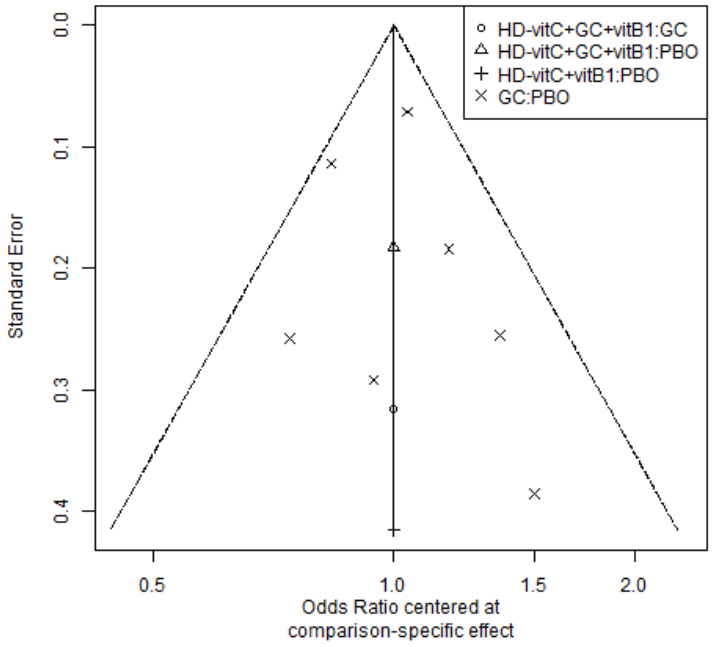
4.3. Risk of bias within studies of each comparison

green bar = low, yellow bar = moderate, red bar = high. GC denotes glucocorticoid; HD-vitC, high dose vitamin C; vitB1, vitamin B1.



4.4. Funnel plot

The treatments were in order of combination therapies first, monotherapies, and placebo or usual care, as the combination therapies have recently been investigated in small trials. GC denotes glucocorticoid; HD-vitC, high dose vitamin C; vitB1, vitamin B1; PBO, placebo or usual care.



4.5. P-scores of treatment

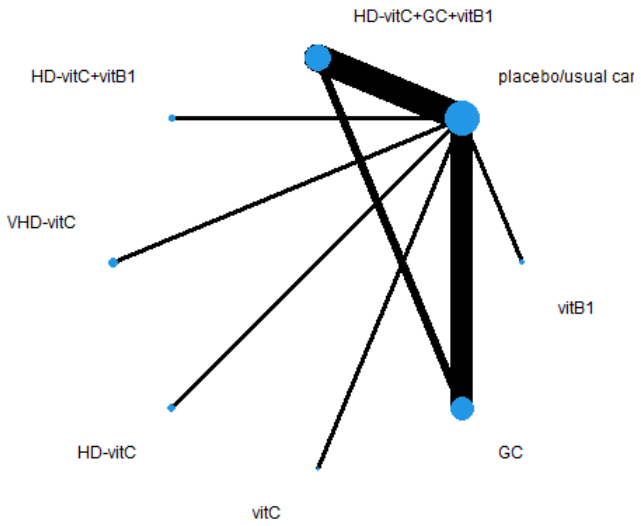
P-scores are calculated from the point estimates and standard errors of the network estimates. The P-score of treatment can be interpreted as the mean extent of certainty that the treatment is better than any other treatment. GC denotes glucocorticoid; HD-vitC, high dose vitamin C; vitB1, vitamin B1.

Treatment	P-score
1. GC	0.8072
2. placebo or usual care	0.5497
3. HD-vitC+vitB1	0.3286
4. HD-vitC+GC+vitB1	0.3146

5. Severity of organ dysfunction over 72 hours

5.1. Network graph

Network graph of all available pairwise comparisons between the eligible interventions. The size of the nodes shows the total number of patients accumulated for each treatment. The breadth of the edges was weighted according to the inverse of the variance of the direct summary effect. GC denotes glucocorticoid; HD-vitC, high dose vitamin C; vitB1, vitamin B1: VHD-vitC, very high dose vitamin C; vitC, vitamin C.



5.2. Heterogeneity: Between-study variance within a comparison and the network

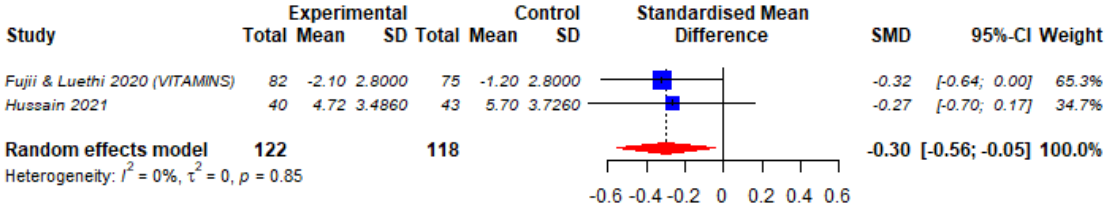
GC denotes glucocorticoid; HD-vitC, high dose vitamin C; vitB1, vitamin B1: VHD-vitC, very high dose vitamin C; vitC, vitamin C; SMD, standardised mean difference; N/A, not available.

Comparisons	Number of studies	Expected variance of SMD from empirical distribution[48] median [95% range]	Estimated variance of SMD		Forest plot
			in comparison	in network	
HD-vitC + GC + vitB1 vs GC	2	0.027 [0.00001 to 4.95]	0	0.0759	No evidence
HD-vitC + GC + vitB1 vs placebo or usual care	6	0.033 [0.0001 to 10.2]	0	0.0759	No heterogeneity
HD-vitC+vitB1 vs placebo or usual care	1	0.033 [0.0001 to 10.2]	N/A	0.0759	N/A
VHD-vitC vs placebo or usual care	1	0.033 [0.0001 to 10.2]	N/A	0.0759	N/A
HD-vitC vs placebo or usual care	1	0.033 [0.0001 to 10.2]	N/A	0.0759	N/A
vitC vs placebo or usual care	1	0.033 [0.0001 to 10.2]	N/A	0.0759	N/A
GC vs placebo or usual care	5	0.033 [0.0001 to 10.2]	0.1740	0.0759	Suspected Heterogeneity
vitB1 vs placebo or usual care	1	0.033 [0.0001 to 10.2]	N/A	0.0759	N/A

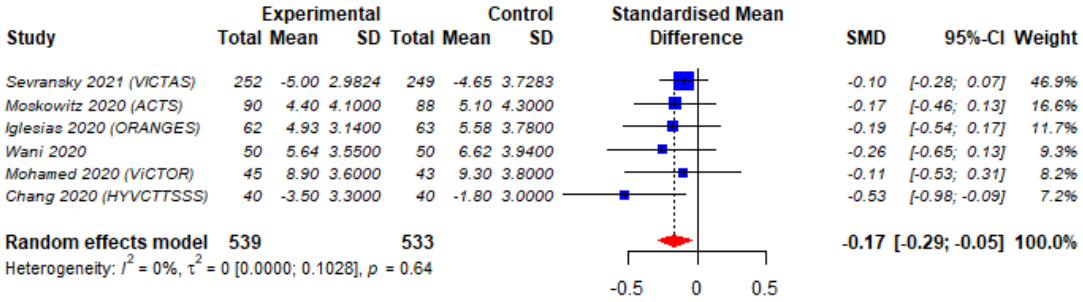
Supplement to: Fujii T, Salanti G, Belletti A, et al. Effect of adjunctive vitamin C, glucocorticoids, and vitamin B1 on longer term mortality in adults with sepsis or septic shock: a systematic review and a component network meta-analysis.

Forest plots

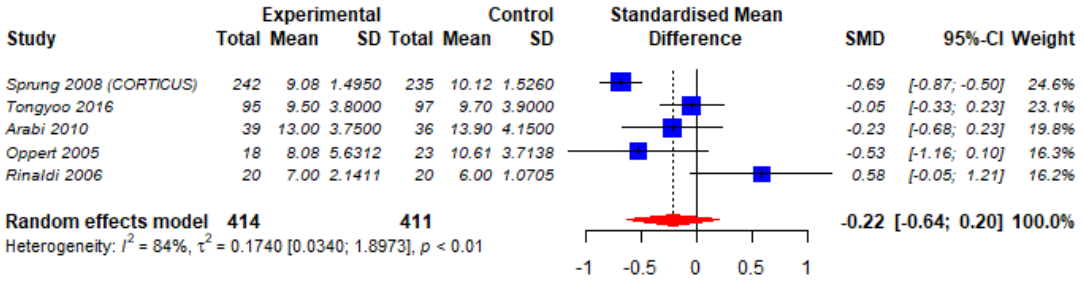
HD-vitC + GC + vitB1 vs GC



HD-vitC + GC + vitB1 vs placebo or usual care



GC vs placebo or usual care



5.3. Inconsistency

5.3.1. side-split

Number of loops = 1. GC denotes glucocorticoid; HD-vitC, high dose vitamin C; vitB1, vitamin B1: VHD-vitC, very high dose vitamin C; vitC, vitamin C; SMD, standardised mean difference.

comparison	Number of studies	SMD from NMA	SMD from direct comparison	SMD from indirect comparison	Difference	p-value
HD-vitC+GC+vitB1 vs HD-vitC+vitB1	0	-0.27	.	-0.27	.	.
HD-vitC+GC+vitB1 vs placebo/usual care	6	-0.27	-0.21	-0.54	0.33	0.2913
HD-vitC+GC+vitB1 vs VHD-vitC	0	-0.55	.	-0.55	.	.
HD-vitC+GC+vitB1 vs vitB1	0	-0.42	.	-0.42	.	.
HD-vitC+GC+vitB1 vs vitC	0	-0.11	.	-0.11	.	.
HD-vitC+vitB1 vs placebo/usual care	1	0	0	.	.	.
HD-vitC+vitB1 vs VHD-vitC	0	-0.27	.	-0.27	.	.
HD-vitC+vitB1 vs vitB1	0	-0.15	.	-0.15	.	.
HD-vitC+vitB1 vs vitC	0	0.17	.	0.17	.	.
VHD-vitC vs placebo/usual care	1	0.27	0.27	.	.	.
VHD-vitC vs vitB1	0	0.12	.	0.12	.	.
VHD-vitC vs vitC	0	0.44	.	0.44	.	.
HD-vitC vs HD-vitC+GC+vitB1	0	-0.15	.	-0.15	.	.
HD-vitC vs HD-vitC+vitB1	0	-0.43	.	-0.43	.	.
HD-vitC vs placebo/usual care	1	-0.43	-0.43	.	.	.
HD-vitC vs VHD-vitC	0	-0.70	.	-0.70	.	.
HD-vitC vs vitB1	0	-0.58	.	-0.58	.	.
HD-vitC vs vitC	0	-0.26	.	-0.26	.	.
vitC vs placebo/usual care	1	-0.17	-0.17	.	.	.
GC vs HD-vitC+GC+vitB1	2	0.11	0.30	-0.03	0.33	0.2913
GC vs HD-vitC+vitB1	0	-0.17	.	-0.17	.	.
GC vs placebo/usual care	5	-0.17	-0.25	0.08	-0.33	0.2913
GC vs VHD-vitC	0	-0.44	.	-0.44	.	.
GC vs HD-vitC	0	0.26	.	0.26	.	.
GC vs vitB1	0	-0.31	.	-0.31	.	.
GC vs vitC	0	0	.	0	.	.
vitB1 vs vitC	0	0.31	.	0.31	.	.
vitB1 vs placebo/usual care	1	0.15	0.15	.	.	.

5.3.2. global

Q statistic to assess consistency under the assumption of a full design-by-treatment interaction random effects model = 1.55 (df=1, p=0.2129)

6. League table for comparisons of all-cause mortality at the longest follow-up in studies published in 2010 or after and severity of organ dysfunction over 72 hours.

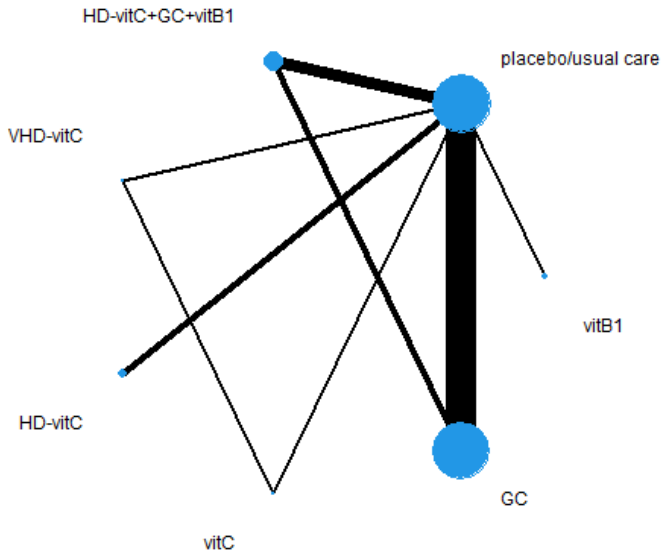
SMD\OR 95% CI 95% PrI	All-cause mortality at the longest follow-up in studies published in 2010 or after (OR) treatment in the column vs treatment in the row							
severity of organ dysfunction over 72 hours (SMD) treatment in the row vs treatment in the column	**placebo or usual care**	1.05 (0.84 to 1.31) (0.83 to 1.33)	1.24 (0.55 to 2.80) (0.52 to 2.95)	**0.50** **(0.27 to 0.92)** **(0.26 to 0.95)**	**0.34** **(0.17 to 0.71)** **(0.16 to 0.74)**	0.33 (0.06 to 1.93) (0.05 to 2.16)	0.97 (0.88 to 1.08) (0.87 to 1.09)	1.1 (0.62 to 1.97) (0.60 to 2.04)
	-0.27 **(-0.51 to -0.04)** (-0.94 to 0.40)	**HD-vitC+GC +vitB1**	1.18 (0.51 to 2.75) (0.48 to 2.90)	**0.48** **(0.25 to 0.91)** **(0.24 to 0.95)**	**0.33** **(0.15 to 0.70)** **(0.15 to 0.73)**	0.31 (0.05 to 1.86) (0.05 to 2.09)	0.93 (0.73 to 1.17) (0.72 to 1.19)	1.05 (0.56 to 1.95) (0.54 to 2.03)
	0 (-0.66 to 0.66) (-0.97 to 0.97)	0.27 (-0.42 to 0.97) (-0.73 to 1.28)	**HD-vitC+vitB1**	0.4 (0.15 to 1.11) (0.14 to 1.19)	**0.28** **(0.09 to 0.82)** **(0.09 to 0.88)**	0.26 (0.04 to 1.86) (0.03 to 2.11)	0.78 (0.34 to 1.78) (0.33 to 1.88)	0.89 (0.33 to 2.42) (0.31 to 2.58)
	0.27 (-0.35 to 0.89) (-0.66 to 1.21)	0.55 (-0.12 to 1.21) (-0.43 to 1.52)	0.27 (-0.63 to 1.18) (-0.92 to 1.47)	**VHD-vitC**	0.69 (0.27 to 1.77) (0.25 to 1.88)	0.66 (0.11 to 3.84) (0.10 to 4.30)	**1.94** **(1.05 to 3.60)** **(1.01 to 3.75)**	2.21 (0.95 to 5.11) (0.90 to 5.39)
	-0.43 (-1.08 to 0.23) (-1.39 to 0.54)	-0.15 (-0.85 to 0.54) (-1.15 to 0.85)	-0.43 (-1.35 to 0.50) (-1.65 to 0.79)	-0.70 (-1.60 to 0.20) (-1.89 to 0.49)	**HD-vitC**	0.95 (0.14 to 6.47) (0.12 to 7.32)	**2.83** **(1.36 to 5.89)** **(1.30 to 6.17)**	**3.22** **(1.27 to 8.13)** **(1.20 to 8.63)**
	-0.17 (-1.04 to 0.71) (-1.33 to 1.00)	0.11 (-0.80 to 1.01) (-1.09 to 1.30)	-0.17 (-1.26 to 0.92) (-1.55 to 1.22)	-0.44 (-1.51 to 0.63) (-1.80 to 0.92)	0.26 (-0.83 to 1.35) (-1.12 to 1.64)	**vitC**	2.97 (0.50 to 17.50) (0.45 to 19.61)	3.37 (0.52 to 21.72) (0.46 to 24.48)
	-0.17 (-0.43 to 0.10) (-0.85 to 0.52)	0.11 (-0.2 to 0.41) (-0.6 to 0.81)	-0.17 (-0.87 to 0.54) (-1.18 to 0.85)	-0.44 (-1.11 to 0.24) (-1.42 to 0.54)	0.26 (-0.44 to 0.97) (-0.75 to 1.27)	0 (-0.91 to 0.91) (-1.20 to 1.20)	**GC**	1.14 (0.63 to 2.04) (0.61 to 2.12)
	0.15 (-0.63 to 0.92) (-0.92 to 1.22)	0.42 (-0.39 to 1.23) (-0.68 to 1.53)	0.15 (-0.87 to 1.16) (-1.16 to 1.46)	-0.12 (-1.12 to 0.87) (-1.41 to 1.16)	0.58 (-0.44 to 1.59) (-0.73 to 1.88)	0.31 (-0.85 to 1.48) (-1.15 to 1.78)	0.31 (-0.5 to 1.13) (-0.8 to 1.43)	**vitB1**

Odds ratios in orange cells denote comparisons of all-cause mortality from 90 days to one year post randomization in studies published in 2010 or after with a treatment in the column versus a treatment in the row. Standardized mean differences in blue cells denote comparisons of severity of organ dysfunction over 72 hours with a treatment in the row versus a treatment in the column. 95% confidence intervals and 95% prediction intervals are presented below the odds ratios. Bold text denotes comparisons where 95% confidence intervals or 95% prediction intervals do not cross null effect (1 for ORs, 0 for SMDs).OR denotes odds ratio; SMD, standardized mean difference; CI, confidence interval; PrI, prediction interval; VHD-vitC, very high-dose vitamin C (≥ 12g per day); HD-vitC, high-dose vitamin C (≥ 6g per day); vitC, vitamin C (< 6g per day); GC, low-dose glucocorticoid (< 400 mg/day); vitB1, vitamin B1 (any dose).

7. Duration of vasopressor therapy

7.1. Network graph

Network graph of all available pairwise comparisons between the eligible interventions. The size of the nodes shows the total number of patients accumulated for each treatment. The breadth of the edges was weighted according to the inverse of the variance of the direct summary effect. GC denotes glucocorticoid; HD-vitC, high dose vitamin C; vitB1, vitamin B1: VHD-vitC, very high dose vitamin C; vitC, vitamin C.



7.2. Heterogeneity: Between-study variance within a comparison and the network

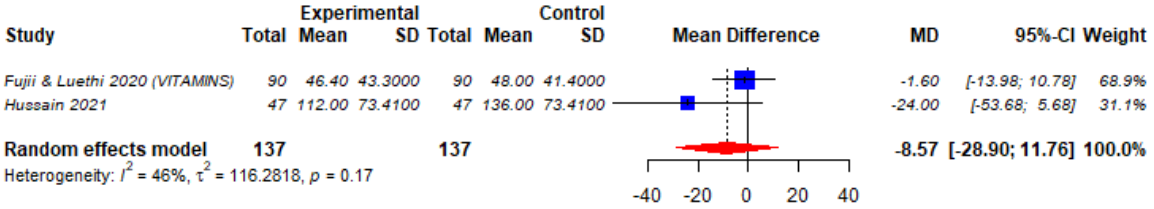
The expected variance from empirical distribution is not available for differences in means of the duration of vasopressor therapy. Standardised mean difference was calculated to compare the estimated variance with the expected variance from empirical distribution. GC denotes glucocorticoid; HD-vitC, high dose vitamin C; vitB1, vitamin B1: VHD-vitC, very high dose vitamin C; vitC, vitamin C; SMD, standardised mean difference; N/A, not available.

Comparisons	Number of studies	Expected variance for SMD from empirical distribution[48] median [95% range]	Estimated variance for SMD	
			in comparison	in network
HD-vitC + GC + vitB1 vs GC	2	0.027 [0.00001 to 4.95]	N/A	0.5013
HD-vitC + GC + vitB1 vs placebo or usual care	4	0.033 [0.0001 to 10.2]	0.0299	0.5013
VHD-vitC vs vitC	1	0.027 [0.00001 to 4.95]	N/A	0.5013
VHD-vitC vs placebo or usual care	1	0.033 [0.0001 to 10.2]	N/A	0.5013
HD-vitC vs placebo or usual care	2	0.033 [0.0001 to 10.2]	0	0.5013
vitC vs placebo or usual care	1	0.033 [0.0001 to 10.2]	N/A	0.5013
vitB1 vs placebo/usual care	1	0.033 [0.0001 to 10.2]	N/A	0.5013
GC vs placebo or usual care	10	0.033 [0.0001 to 10.2]	0.6336	0.5013

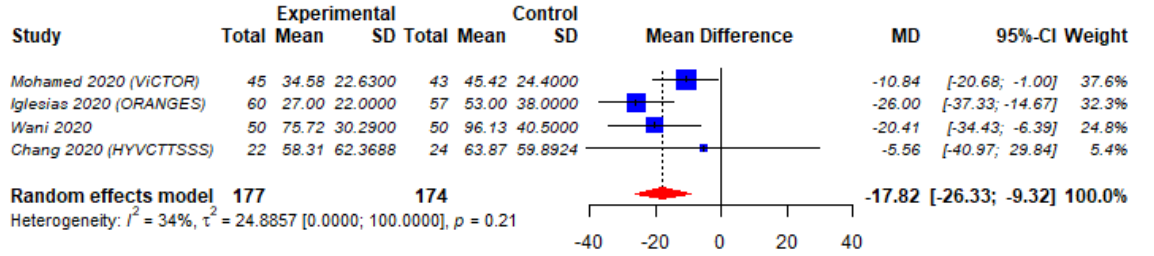
Supplement to: Fujii T, Salanti G, Belletti A, et al. Effect of adjunctive vitamin C, glucocorticoids, and vitamin B1 on longer term mortality in adults with sepsis or septic shock: a systematic review and a component network meta-analysis.

Forest plots

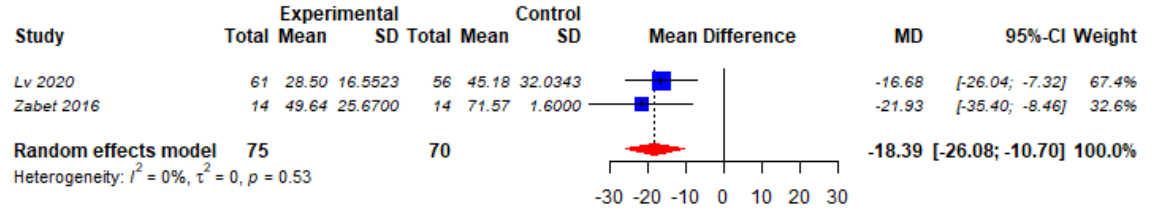
HD-vitC + GC + vitB1 vs GC



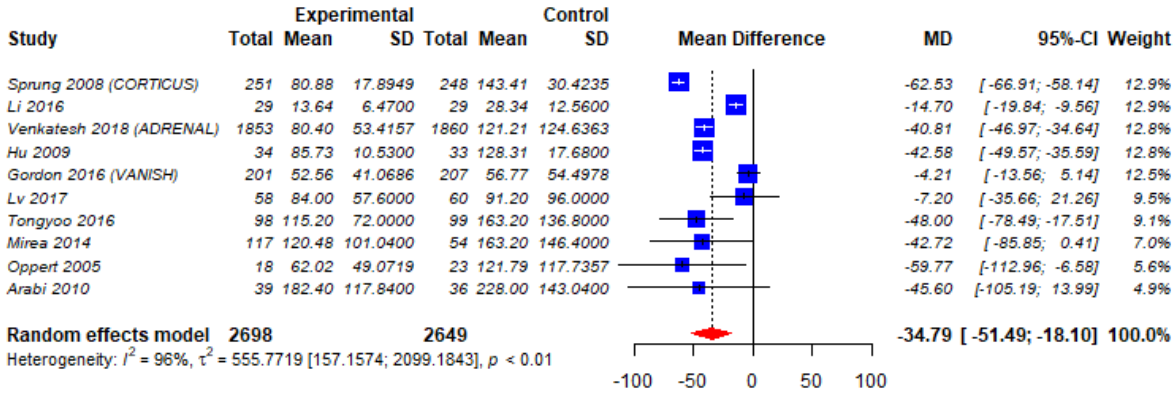
HD-vitC + GC + vitB1vs placebo or usual care



HD-vitC vs placebo or usual care



GC vs placebo or usual care



7.3. Inconsistency

7.3.1. Side-split

Number of loops = 2. GC denotes glucocorticoid; HD-vitC, high dose vitamin C; vitB1, vitamin B1: VHD-vitC, very high dose vitamin C; vitC, vitamin C; MD, mean difference; N/A, not available.

comparison	Number of studies	MD from NMA	MD from direct comparison	MD from indirect comparison	Difference	p-value
GC vs HD-vitC	0	-11.71	.	-11.71	.	.
GC vs HD-vitC+GC+vitB1	2	-6.16	11.02	-17.95	28.97	0.1904
GC vs placebo/usual care	10	-30.95	-34.63	-5.66	-28.97	0.1904
GC vs VHD-vitC	0	-23.75	.	-23.75	.	.
GC vs vitB1	0	-21.55	.	-21.55	.	.
GC vs vitC	0	12.25	.	12.25	.	.
HD-vitC vs HD-vitC+GC+vitB1	0	5.55	.	5.55	.	.
HD-vitC vs placebo/usual care	2	-19.24	-19.24	.	.	.
HD-vitC vs VHD-vitC	0	-12.04	.	-12.04	.	.
HD-vitC vs vitB1	0	-9.84	.	-9.84	.	.
HD-vitC vs vitC	0	23.96	.	23.96	.	.
HD-vitC+GC+vitB1 vs placebo/usual care	4	-24.79	-16.68	-45.65	28.97	0.1904
HD-vitC+GC+vitB1 vs VHD-vitC	0	-17.59	.	-17.59	.	.
HD-vitC+GC+vitB1 vs vitB1	0	-15.39	.	-15.39	.	.
HD-vitC+GC+vitB1 vs vitC	0	18.41	.	18.41	.	.

7.3.2. global

Q statistic to assess consistency under the assumption of a full design-by-treatment interaction random effects model = 1.72 (df=1, p=0.1899)

8. League table for comparisons of time to cessation of vasopressor therapy and ICU length of stay.

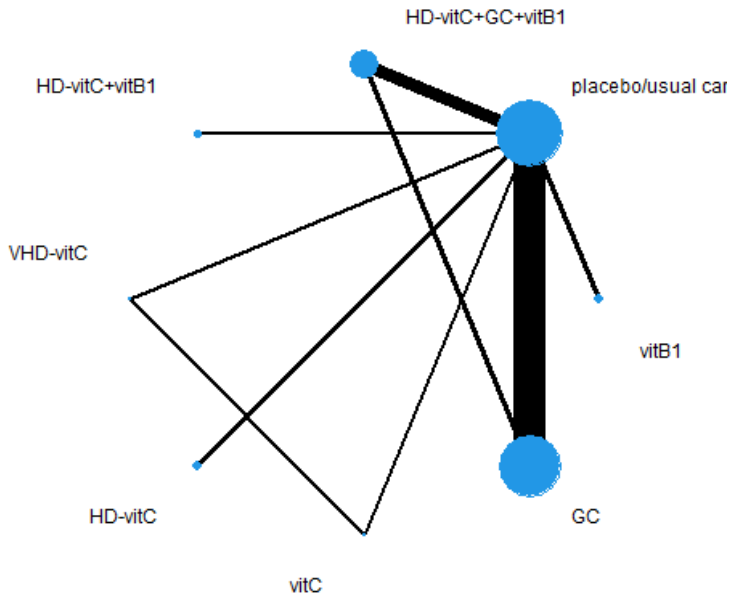
MD 95%CI 95%PrI	Time to cessation of vasopressor therapy, hours treatment in the column vs treatment in the row							
ICU length of stay, days treatment in the row vs treatment in the column	**placebo or usual care**	**-24.8 (-44.3 to -5.3) **(-2.3 to 1.0)	No data available	-7.2 (-74.9 to 60.5) (-2.5 to 2.3)	-19.2 (-50.1 to 11.6) (-2.8 to 1.0)	-43.2 (-110.9 to 24.5) (-3.2 to 1.7)	**-31.0 (-45.4 to -16.5) **(-2.4 to 0.8)	-9.4 (-63.8 to 45.0) (-2.3 to 2.1)
0.3 (-0.9 to 1.5) (-2.4 to 3.0)	**HD-vitC+GC +vitB1**	No data available	17.6 (-52.8 to 88) (-1.9 to 3.0)	5.5 (-30.9 to 42.0) (-2.2 to 1.8)	-18.4 (-88.8 to 52.0) (-2.6 to 2.4)	-6.2 (-27.5 to 15.1) (-1.8 to 1.5)	15.4 (-42.4 to 73.2) (-1.8 to 2.8)	
-1 (-4.3 to 2.3) (-5.3 to 3.2)	-1.3 (-4.8 to 2.2) (-5.8 to 3.1)	**HD-vitC+vitB1**	No data available	No data available	No data available	No data available	No data available	
-1.9 (-12.4 to 8.6) (-13.4 to 9.6)	-2.2 (-12.8 to 8.4) (-13.8 to 9.4)	-0.9 (-11.9 to 10.1) (-12.9 to 11.2)	**VHD-vitC**	-12.0 (-86.4 to 62.3) (-3.4 to 1.9)	-36.0 (-103.7 to 31.7) (-3.0 to 1.8)	-23.8 (-93.0 to 45.5) (-3.1 to 1.7)	-2.2 (-89.0 to 84.6) (-2.9 to 2.9)	
0.4 (-2.0 to 2.8) (-3.1 to 3.9)	0.1 (-2.6 to 2.8) (-3.6 to 3.9)	1.4 (-2.7 to 5.5) (-3.5 to 6.4)	2.3 (-8.5 to 13.1) (-9.5 to 14.1)	**HD-vitC**	-24.0 (-98.3 to 50.4) (-2.6 to 2.8)	-11.7 (-45.8 to 22.4) (-1.9 to 2.0)	9.8 (-52.7 to 72.4) (-1.7 to 3.3)	
-2.9 (-13.4 to 7.6) (-14.4 to 8.6)	-3.2 (-13.8 to 7.4) (-14.8 to 8.4)	-1.9 (-12.9 to 9.1) (-13.9 to 10.2)	-1.0 (-11.5 to 9.5) (-12.5 to 10.5)	-3.3 (-14.1 to 7.5) (-15.1 to 8.5)	**vitC**	12.2 (-57.0 to 81.5) (-2.5 to 2.4)	33.8 (-53.0 to 120.6) (-2.3 to 3.5)	
**-1.5 (-2.4 to -0.5) **(-4.0 to 1.1)	**-1.8 (-3.1 to -0.4) **(-4.5 to 1.0)	-0.4 (-3.9 to 3.0) (-4.8 to 3.9)	0.4 (-10.1 to 11.0) (-11.2 to 12.0)	-1.9 (-4.5 to 0.7) (-5.5 to 1.8)	1.4 (-9.1 to 12.0) (-10.2 to 13.0)	**GC**	21.6 (-34.7 to 77.8) (-1.6 to 3.0)	
-0.7 (-3.5 to 2.2) (-4.5 to 3.2)	-0.9 (-4.1 to 2.2) (-5.0 to 3.1)	0.4 (-4.0 to 4.7) (-4.9 to 5.6)	1.2 (-9.6 to 12.1) (-10.7 to 13.2)	-1.1 (-4.8 to 2.7) (-5.7 to 3.6)	2.2 (-8.6 to 13.1) (-9.7 to 14.2)	0.8 (-2.2 to 3.8) (-3.2 to 4.8)	**vitB1**	

Mean differences in orange cells denote comparisons of time to cessation of vasopressor therapy with a treatment in the column versus a treatment in the row. Mean differences in blue cells denote comparisons of ICU length of stay with a treatment in the row versus a treatment in the column. 95% confidence intervals and 95% prediction intervals are presented below the odds ratios. Bold text denotes comparisons where 95% confidence intervals or 95% prediction intervals do not cross 0. MD denotes mean difference; CI, confidence interval; PrI, predictive interval; VHD-vitC, very high-dose vitamin C (≥ 12g per day); HD-vitC, high-dose vitamin C (≥ 6g per day); vitC, vitamin C (< 6g per day); GC, low-dose glucocorticoid (< 400 mg/day); vitB1, vitamin B1 (any dose).

9. ICU length of stay

9.1. Network graph

Network graph of all available pairwise comparisons between the eligible interventions. The size of the nodes shows the total number of patients accumulated for each treatment. The breadth of the edges was weighted according to the inverse of the variance of the direct summary effect. GC denotes glucocorticoid; HD-vitC, high dose vitamin C; vitB1, vitamin B1: VHD-vitC, very high dose vitamin C; vitC, vitamin C.



9.2. Heterogeneity: Between-study variance within a comparison and the network

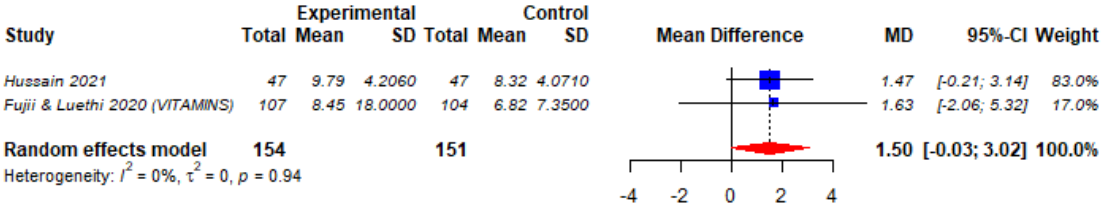
The expected variance from empirical distribution is not available for differences in means of the length of ICU stay. Standardised mean difference was calculated to compare the estimated variance with the expected variance from empirical distribution. GC denotes glucocorticoid; HD-vitC, high dose vitamin C; vitB1, vitamin B1: VHD-vitC, very high dose vitamin C; vitC, vitamin C; SMD, standardised mean difference; N/A, not available.

Comparisons	Number of studies	Expected variance of SMD from empirical distribution[48] median [95% range]	Estimated variance of SMD	
			in comparison	in network
HD-vitC + GC + vitB1 vs GC	2	0.027 [0.00001 to 4.95]	0	1.2155
HD-vitC + GC + vitB1 vs placebo or usual care	5	0.033 [0.0001 to 10.2]	0	1.2155
VHD-vitC vs placebo or usual care	1	0.033 [0.0001 to 10.2]	N/A	1.2155
HD-vitC vs placebo or usual care	2	0.033 [0.0001 to 10.2]	0	1.2155
vitC vs placebo or usual care	1	0.033 [0.0001 to 10.2]	N/A	1.2155
GC vs placebo or usual care	11	0.033 [0.0001 to 10.2]	0.0071	1.2155
vitB1 vs placebo or usual care	2	0.033 [0.0001 to 10.2]	0	1.2155

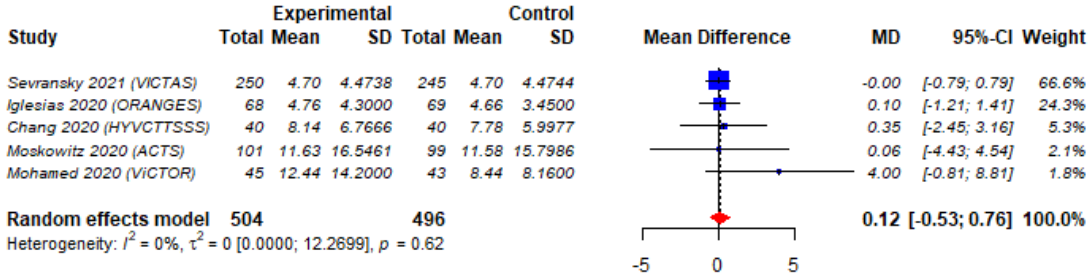
Supplement to: Fujii T, Salanti G, Belletti A, et al. Effect of adjunctive vitamin C, glucocorticoids, and vitamin B1 on longer term mortality in adults with sepsis or septic shock: a systematic review and a component network meta-analysis.

Forest plots

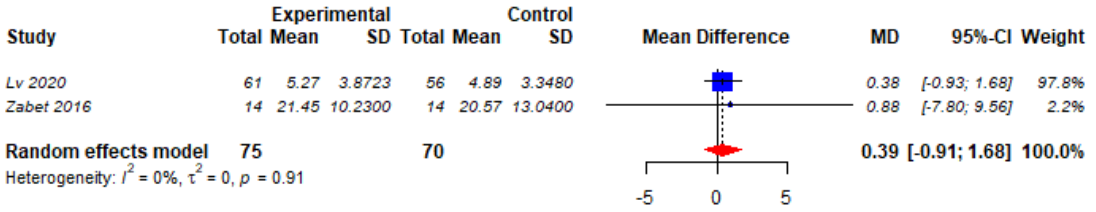
HD-vitC + GC + vitB1 vs GC



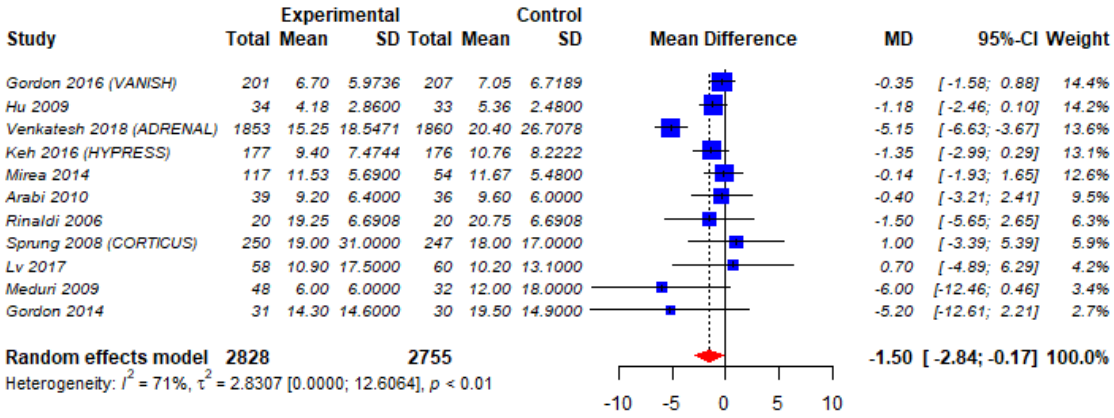
HD-vitC + GC + vitB1 vs placebo or usual care



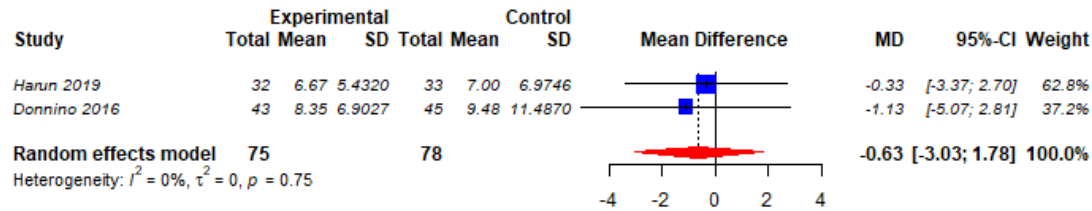
HD-vitC vs placebo or usual care



GC vs placebo or usual care



vitB1 vs placebo or usual care



9.3. Inconsistency

9.3.1. side-split

Number of loops = 2. GC denotes glucocorticoid; HD-vitC, high dose vitamin C; vitB1, vitamin B1: VHD-vitC, very high dose vitamin C; vitC, vitamin C; MD, mean difference.

comparison	Number of studies	MD from NMA	MD from direct comparison	MD from indirect comparison	Difference	p-value
GC vs HD-vitC	0	-1.88	.	-1.88	.	.
GC vs HD-vitC+GC+vitB1	2	-1.75	-1.51	-1.89	0.37	0.8003
GC vs HD-vitC+vitB1	0	-0.43	.	-0.43	.	.
GC vs placebo/usual care	11	-1.46	-1.51	-1.14	-0.37	0.8003
GC vs VHD-vitC	0	0.44	.	0.44	.	.
GC vs vitB1	0	-0.81	.	-0.81	.	.
GC vs vitC	0	1.44	.	1.44	.	.
HD-vitC vs HD-vitC+GC+vitB1	0	0.12	.	0.12	.	.
HD-vitC vs HD-vitC+vitB1	0	1.44	.	1.44	.	.
HD-vitC vs placebo/usual care	2	0.41	0.41	.	.	.
HD-vitC vs VHD-vitC	0	2.31	.	2.31	.	.
HD-vitC vs vitB1	0	1.07	.	1.07	.	.
HD-vitC vs vitC	0	3.31	.	3.31	.	.
HD-vitC+GC+vitB1 vs HD-vitC+vitB1	0	1.32	.	1.32	.	.
HD-vitC+GC+vitB1 vs placebo/usual care	5	0.29	0.38	0.01	0.37	0.8003
HD-vitC+GC+vitB1 vs VHD-vitC	0	2.19	.	2.19	.	.
HD-vitC+GC+vitB1 vs vitB1	0	0.95	.	0.95	.	.
HD-vitC+GC+vitB1 vs vitC	0	3.19	.	3.19	.	.
HD-vitC+vitB1 vs placebo/usual care	1	-1.03	-1.03	.	.	.
HD-vitC+vitB1 vs VHD-vitC	0	0.87	.	0.87	.	.
HD-vitC+vitB1 vs vitB1	0	-0.37	.	-0.37	.	.
HD-vitC+vitB1 vs vitC	0	1.87	.	1.87	.	.
VHD-vitC vs placebo/usual care	1	-1.9	-1.9	.	.	.
vitB1 vs placebo/usual care	2	-0.66	-0.66	.	.	.
vitC vs placebo/usual care	1	-2.9	-2.9	.	.	.
VHD-vitC vs vitB1	0	-1.24	.	-1.24	.	.
VHD-vitC vs vitC	1	1	1	.	.	.
vitB1 vs vitC	0	2.24	.	2.24	.	.

9.3.2. global

Q statistic to assess consistency under the assumption of a full design-by-treatment interaction random effects model = 0.06 (df=1, p=0.7989)

10. All-cause mortality at the longest follow-up

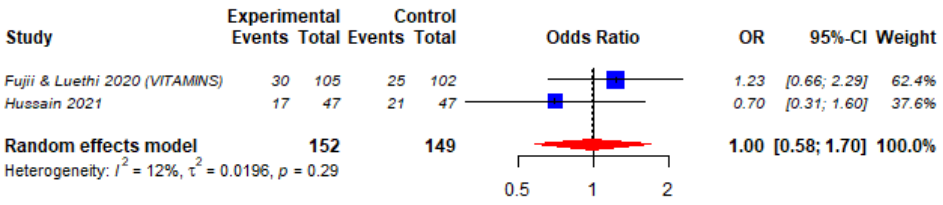
10.1. Heterogeneity: Between-study variance within a comparison and the network

GC denotes glucocorticoid; HD-vitC, high dose vitamin C; vitB1, vitamin B1: VHD-vitC, very high dose vitamin C; vitC, vitamin C; OR, odds ratio; N/A, not available.

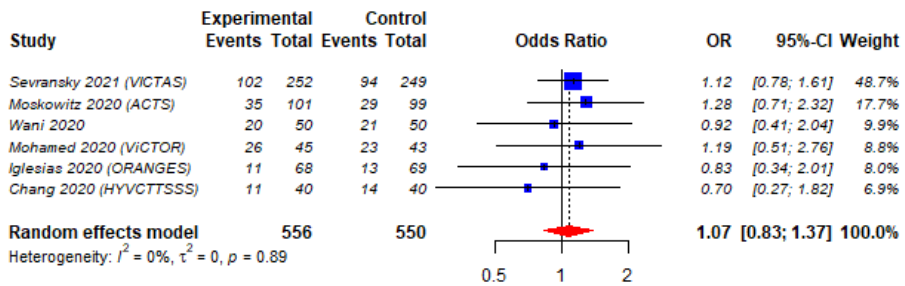
Comparisons	Number of studies	Expected variance for ORs from empirical distribution[47] median [95% range]	Estimated variance		Forest plot
			in comparison	in network	
HD-vitC + GC + vitB1 vs GC	2	0.014 [0.0008 to 0.25]	0.0196	0.0018	No evidence
HD-vitC + GC + vitB1 vs placebo or usual care	6	0.017 [0.001 to 0.30]	0	0.0018	No heterogeneity
HD-vitC+vitB1 vs placebo or usual care	1	0.017 [0.001 to 0.30]	N/A	0.0018	N/A
VHD-vitC vs vitC	1	0.014 [0.0008 to 0.25]	N/A	0.0018	N/A
VHD-vitC vs placebo or usual care	2	0.017 [0.001 to 0.30]	0	0.0018	No evidence
HD-vitC vs placebo or usual care	2	0.017 [0.001 to 0.30]	0.6702	0.0018	No evidence
vitC vs placebo or usual care	2	0.017 [0.001 to 0.30]	0.7293	0.0018	No evidence
GC vs placebo or usual care	17	0.017 [0.001 to 0.30]	0.0239	0.0018	No heterogeneity
vitB1 vs placebo or usual care	3	0.017 [0.001 to 0.30]	0	0.0018	No evidence

Forest plot

HD-vitC + GC + vitB1 vs GC

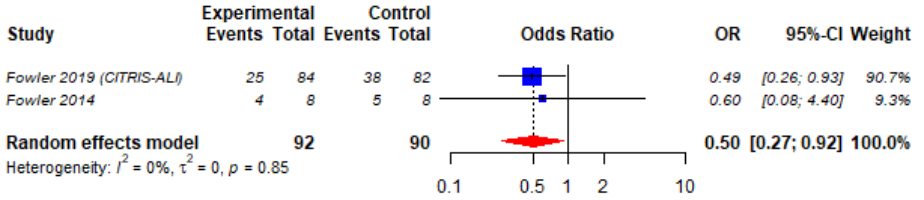


HD-vitC + GC + vitB1 vs placebo or usual care

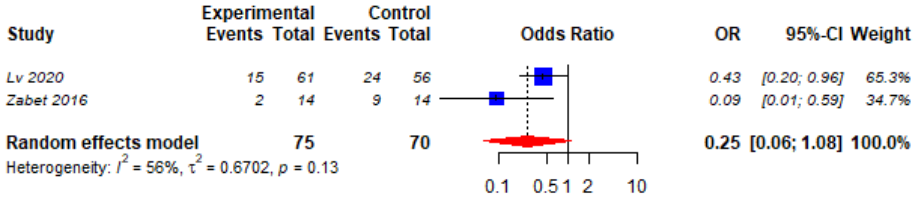


Supplement to: Fujii T, Salanti G, Belletti A, et al. Effect of adjunctive vitamin C, glucocorticoids, and vitamin B1 on longer term mortality in adults with sepsis or septic shock: a systematic review and a component network meta-analysis.

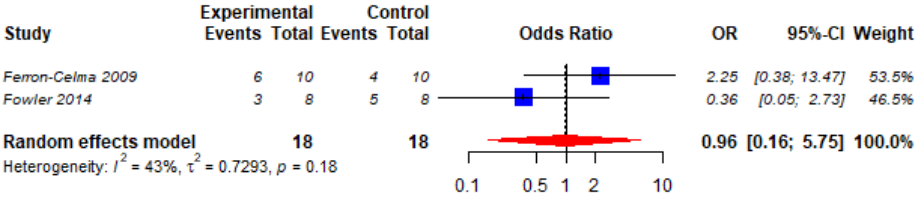
VHD-vitC vs placebo or usual care



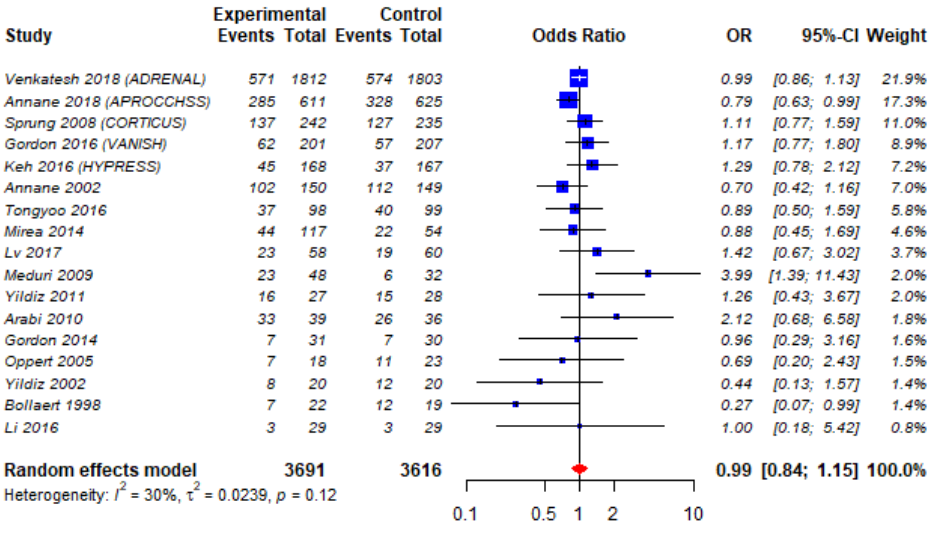
HD-vitC vs placebo or usual care



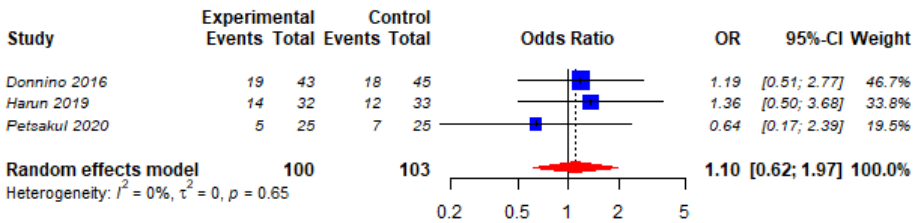
vitC vs placebo or usual care



GC vs placebo or usual care



vitB1 vs placebo or usual care



10.2. Inconsistency

10.2.1. side-split

Number of loops = 2. GC denotes glucocorticoid; HD-vitC, high dose vitamin C; vitB1, vitamin B1: VHD-vitC, very high dose vitamin C; vitC, vitamin C; OR, odds ratio; RoR, Ratio of Ratios (direct versus indirect).

comparison	Number of studies	OR from NMA	OR from direct comparison	OR from indirect comparison	Ratio of ratios	p-value
HD-vitC+GC+vitB1 vs HD-vitC+vitB1	0	0.84	.	0.84	.	.
HD-vitC+GC+vitB1 vs placebo or usual care	6	1.04	1.06	0.98	1.09	0.7772
HD-vitC+GC+vitB1 vs VHD-vitC	0	1.96	.	1.96	.	.
HD-vitC+GC+vitB1 vs vitB1	0	0.95	.	0.95	.	.
HD-vitC+GC+vitB1 vs vitC	0	1.23	.	1.23	.	.
HD-vitC vs HD-vitC+GC+vitB1	0	0.33	.	0.33	.	.
HD-vitC vs HD-vitC+vitB1	0	0.27	.	0.27	.	.
HD-vitC vs VHD-vitC	0	0.64	.	0.64	.	.
HD-vitC vs vitB1	0	0.31	.	0.31	.	.
HD-vitC vs vitC	0	0.40	.	0.40	.	.
HD-vitC+vitB1 vs placebo or usual care	1	1.24	1.24	.	.	.
HD-vitC+vitB1 vs VHD-vitC	0	2.33	.	2.33	.	.
HD-vitC+vitB1 vs vitB1	0	1.13	.	1.13	.	.
HD-vitC+vitB1 vs vitC	0	1.46	.	1.46	.	.
VHD-vitC vs vitB1	0	0.48	.	0.48	.	.
VHD-vitC vs vitC	1	0.63	1.67	0.28	5.86	0.1973
VHD-vitC vs placebo or usual care	2	0.53	0.50	20.92	0.02	0.1358
HD-vitC vs placebo or usual care	2	0.34	0.34	.	.	.
vitC vs placebo or usual care	2	0.85	1.01	0.24	4.18	0.4759
GC vs HD-vitC+GC+vitB1	2	0.94	1.00	0.92	1.09	0.7772
GC vs HD-vitC+vitB1	0	0.79	.	0.79	.	.
GC vs VHD-vitC	0	1.84	.	1.84	.	.
GC vs HD-vitC	0	2.87	.	2.87	.	.
GC vs placebo or usual care	17	0.98	0.97	1.06	0.92	0.7772
GC vs vitB1	0	0.89	.	0.89	.	.
GC vs vitC	0	1.15	.	1.15	.	.
vitB1 vs vitC	0	1.29	.	1.29	.	.
vitB1 vs placebo or usual care	3	1.1	1.1	.	.	.

10.2.2. global

Q statistic to assess consistency under the assumption of a full design-by-treatment interaction random effects model = 2.37 (df=3, p=0.5000)

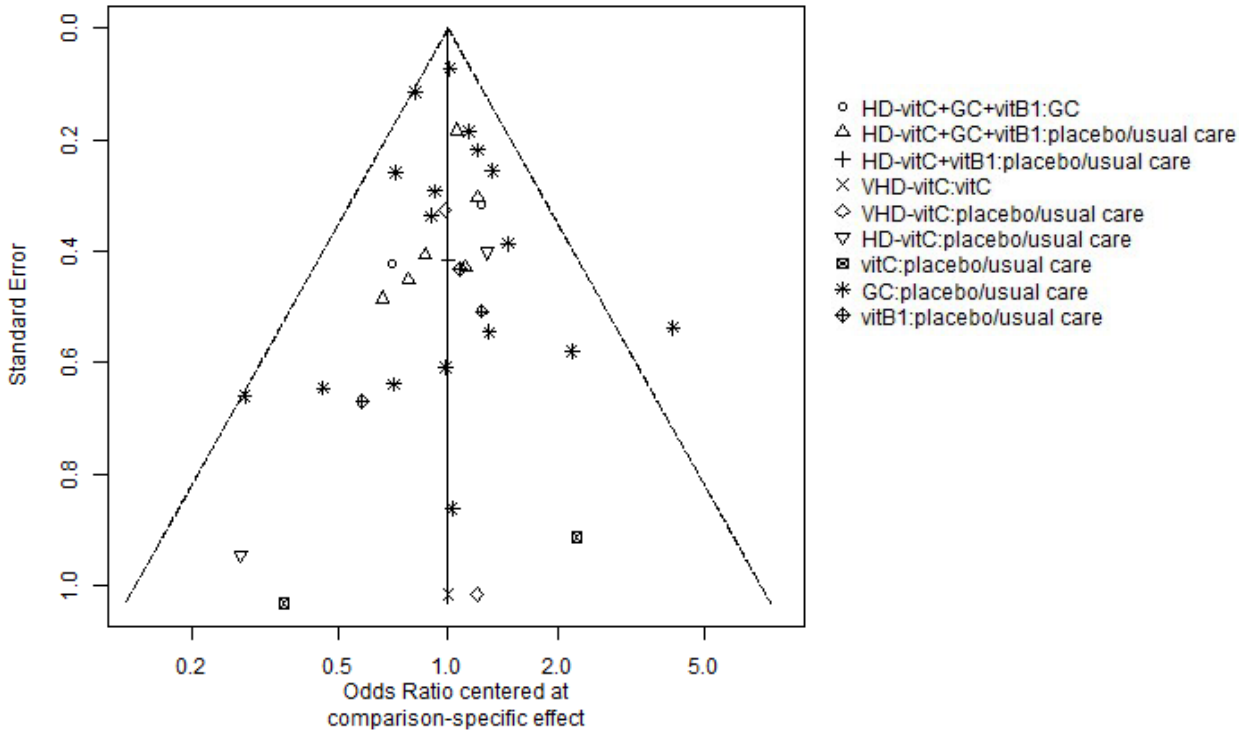
10.3. Risk of bias within studies of each comparison

vitC denotes vitamin C; vitB1, vitamin B1.



10.4. Funnel plot

The treatments were in order of combination therapies first, monotherapies, and placebo or usual care, as the combination therapies have recently been investigated in small trials. GC denotes glucocorticoid; HD-vitC, high dose vitamin C; vitB1, vitamin B1: VHD-vitC, very high dose vitamin C; vitC, vitamin C.



10.5. Summary of confidence in network estimates

Clinically important threshold for odds ratio was defined at 1.2 to judge imprecision of the effect estimates. Incoherence in the summary table refers to inconsistency assessment in 5.2. vitC denotes vitamin C; vitB1, vitamin B1.

Comparison	Number of studies	Within-study bias	Reporting bias	Indirectness	Imprecision	Heterogeneity	Incoherence	Confidence rating
MIXED EVIDENCE								
glucocorticoid vs placebo/usual care	17	Some concerns	Low risk	No concerns	No concerns	No concerns	No concerns	Moderate
placebo/usual care vs vitC (high dose)+glucocorticoid+vitB1	6	Some concerns	Low risk	No concerns	Major concerns	No concerns	No concerns	Very low
placebo/usual care vs vitB1	3	Major concerns	Low risk	No concerns	Major concerns	No concerns	No concerns	Very low
placebo/usual care vs vitC (high dose)	2	Major concerns	Low risk	No concerns	No concerns	No concerns	No concerns	Low
placebo/usual care vs vitC (very high dose)	2	Some concerns	Low risk	No concerns	No concerns	No concerns	Some concerns	Low
glucocorticoid vs vitC (high dose)+glucocorticoid+vitB1	2	Some concerns	Low risk	No concerns	Major concerns	No concerns	No concerns	Very low
placebo/usual care vs vitC	2	Some concerns	Low risk	No concerns	Major concerns	No concerns	Some concerns	Very low
placebo/usual care vs vitC (high dose)+vitB1	1	No concerns	Low risk	No concerns	Major concerns	No concerns	No concerns	Low
vitC vs vitC (very high dose)	1	Some concerns	Low risk	No concerns	Major concerns	No concerns	Some concerns	Very low
INDIRECT EVIDENCE								
glucocorticoid vs vitC (high dose)+vitB1	0	No concerns	Low risk	No concerns	Major concerns	No concerns	No concerns	Low
vitC (high dose)+vitB1 vs vitC (very high dose)	0	No concerns	Low risk	No concerns	Major concerns	No concerns	No concerns	Low
vitB1 vs vitC (high dose)	0	Major concerns	Low risk	No concerns	No concerns	No concerns	No concerns	Low
vitC (high dose) vs vitC (high dose)+glucocorticoid+vitB1	0	Major concerns	Low risk	No concerns	No concerns	No concerns	No concerns	Low

glucocorticoid vs vitC (very high dose)	0	Some concerns	Low risk	No concerns	Some concerns	No concerns	No concerns	Low
vitB1 vs vitC (very high dose)	0	Some concerns	Low risk	No concerns	Some concerns	No concerns	No concerns	Low
glucocorticoid vs vitC (high dose)	0	Some concerns	Low risk	No concerns	No concerns	No concerns	No concerns	Moderate
vitC (high dose) vs vitC (high dose)+vitB1	0	Some concerns	Low risk	No concerns	No concerns	No concerns	No concerns	Moderate
vitC (high dose)+glucocorticoid+vitB1 vs vitC (very high dose)	0	Some concerns	Low risk	No concerns	No concerns	No concerns	No concerns	Moderate
glucocorticoid vs vitB1	0	Some concerns	Low risk	No concerns	Major concerns	No concerns	No concerns	Very low
glucocorticoid vs vitC	0	Some concerns	Low risk	No concerns	Major concerns	No concerns	No concerns	Very low
vitB1 vs vitC	0	Some concerns	Low risk	No concerns	Major concerns	No concerns	No concerns	Very low
vitB1 vs vitC (high dose)+glucocorticoid+vitB1	0	Some concerns	Low risk	No concerns	Major concerns	No concerns	No concerns	Very low
vitB1 vs vitC (high dose)+vitB1	0	Some concerns	Low risk	No concerns	Major concerns	No concerns	No concerns	Very low
vitC vs vitC (high dose)	0	Some concerns	Low risk	No concerns	Major concerns	No concerns	No concerns	Very low
vitC vs vitC (high dose)+glucocorticoid+vitB1	0	Some concerns	Low risk	No concerns	Major concerns	No concerns	No concerns	Very low
vitC vs vitC (high dose)+vitB1	0	Some concerns	Low risk	No concerns	Major concerns	No concerns	No concerns	Very low
vitC (high dose) vs vitC (very high dose)	0	Some concerns	Low risk	No concerns	Major concerns	No concerns	No concerns	Very low
vitC (high dose)+glucocorticoid+vitB1 vs vitC (high dose)+vitB1	0	Some concerns	Low risk	No concerns	Major concerns	No concerns	No concerns	Very low

10.6. P-scores of treatment

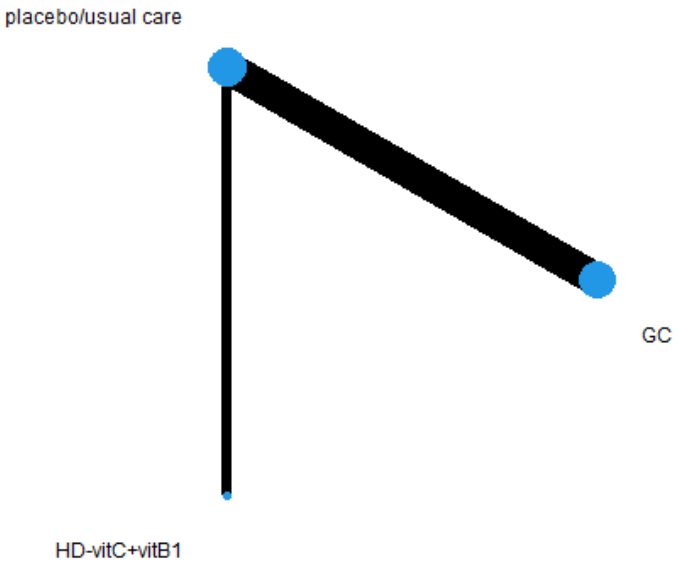
P-scores are calculated from the point estimates and standard errors of the network estimates. The P-score of treatment can be interpreted as the mean extent of certainty that the treatment is better than any other treatment. HD-vitC denotes high dose vitamin C; VHD-vitC, very high dose vitamin C; vitC, vitamin C; GC, glucocorticoid; vitB1, vitamin B1.

Treatment	P-score
1. HD-vitC	0.9543
2. VHD-vitC	0.8219
3. vitC	0.4987
4. GC	0.4482
5. placebo or usual care	0.3927
6. HD-vitC+GC+vitB1	0.3266
7. vitB1	0.3089
8. HD-vitC+vitB1	0.2485

11. All-cause mortality from 90 days to 1 year: only studies with low risk of bias

11.1. Network graph

Network graph of all available pairwise comparisons between the eligible interventions. The size of the nodes shows the total number of patients accumulated for each treatment. The breadth of the edges was weighted according to the inverse of the variance of the direct summary effect. GC denotes glucocorticoid; HD-vitC, high dose vitamin C; vitB1, vitamin B1: VHD-vitC, very high dose vitamin C; vitC, vitamin C.



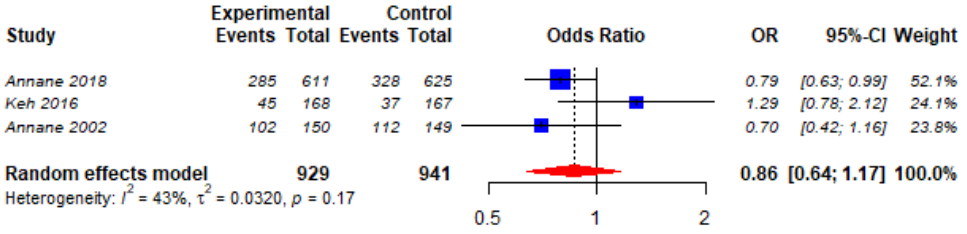
11.2. Heterogeneity: Between-study variance within a comparison and the network

GC denotes glucocorticoid; HD-vitC, high dose vitamin C; vitB1, vitamin B1: VHD-vitC, very high dose vitamin C; vitC, vitamin C; OR, odds ratio; N/A, not available.

Comparisons	Number of studies	Expected variance for ORs from empirical distribution[47] median [95% range]	Estimated variance		Forest plot
			in comparison	in network	
HD-vitC+vitB1 vs placebo or usual care	1	0.017 [0.001 to 0.30]	N/A	0.0320	N/A
GC vs placebo or usual care	3	0.017 [0.001 to 0.30]	0.0320	0.0320	Suspected heterogeneity

Forest plot

GC vs placebo or usual care



11.3. League table of NMA results

Odds ratios denote comparisons of all-cause mortality from 90 days to one year in low risk of bias studies with a treatment in the row versus a treatment in the column. 95% confidence intervals and 95% prediction intervals are presented below the odds ratios. GC denotes glucocorticoid; HD-vitC, high dose vitamin C; vitB1, vitamin B1.

OR **(95% CI)** **(95% PrI)**	**placebo or usual care**	**HD-vitC + vitB1**
HD-vitC + vitB1	1.24 (0.51 to 3.01) (0.003 to 601.3)	
GC	0.86 (0.64 to 1.17) (0.04 to 17.22)	0.70 (0.27 to 1.78) (0.001 to 456.22)

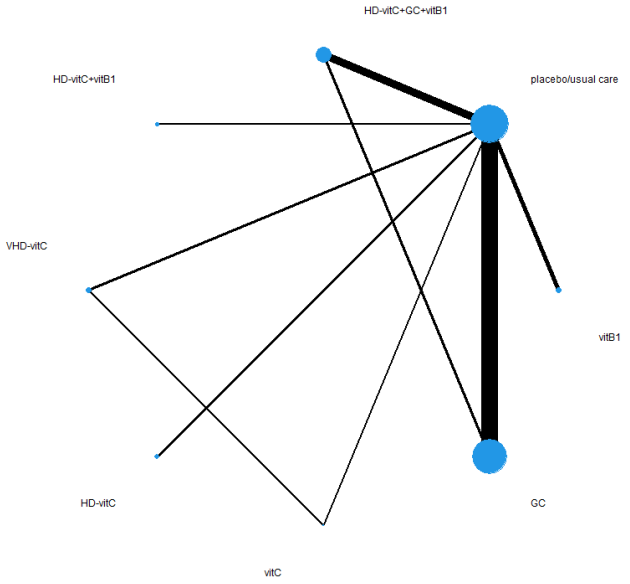
11.4. Component network meta-analysis

Component network meta-analysis was not performed due to the scarcity of available data.

12. All-cause mortality at the longest follow-up: Only studies published in 2010 or after

12.1. Network graph

Network graph of all available pairwise comparisons between the eligible interventions. The size of the nodes shows the total number of patients accumulated for each treatment. The breadth of the edges was weighted according to the inverse of the variance of the direct summary effect. GC denotes glucocorticoid; HD-vitC, high dose vitamin C; vitB1, vitamin B1: VHD-vitC, very high dose vitamin C; vitC, vitamin C.



12.2. Heterogeneity: Between-study variance within a comparison and the network

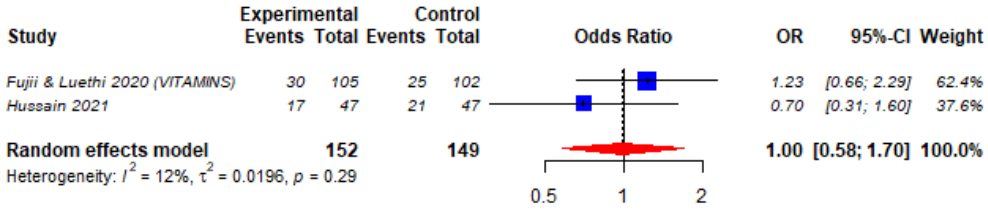
GC denotes glucocorticoid; HD-vitC, high dose vitamin C; vitB1, vitamin B1: VHD-vitC, very high dose vitamin C; vitC, vitamin C; OR, odds ratio; N/A, not available.

Comparisons	Number of studies	Expected variance for ORs from empirical distribution[47] median [95% range]	Estimated variance		Forest plot
			in comparison	in network	
HD-vitC + GC + vitB1 vs GC	2	0.014 [0.0008 to 0.25]	0.0196	0	No evidence
HD-vitC + GC + vitB1 vs placebo or usual care	6	0.017 [0.001 to 0.30]	0	0	No heterogeneity
HD-vitC+vitB1 vs placebo or usual care	1	0.017 [0.001 to 0.30]	N/A	0	N/A
VHD-vitC vs vitC	1	0.014 [0.0008 to 0.25]	N/A	0	N/A
VHD-vitC vs placebo or usual care	2	0.017 [0.001 to 0.30]	0	0	No evidence
HD-vitC vs placebo or usual care	2	0.017 [0.001 to 0.30]	0.6702	0	No evidence
vitC vs placebo or usual care	1	0.017 [0.001 to 0.30]	N/A	0	N/A
GC vs placebo or usual care	11	0.017 [0.001 to 0.30]	0	0	No heterogeneity
vitB1 vs placebo or usual care	3	0.017 [0.001 to 0.30]	0	0	No evidence

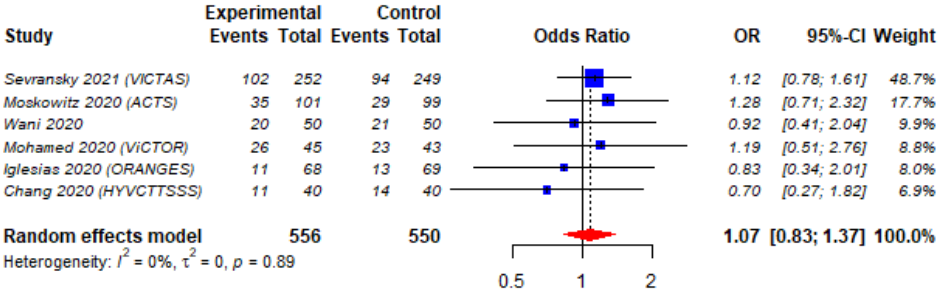
Supplement to: Fujii T, Salanti G, Belletti A, et al. Effect of adjunctive vitamin C, glucocorticoids, and vitamin B1 on longer term mortality in adults with sepsis or septic shock: a systematic review and a component network meta-analysis.

Forest plots

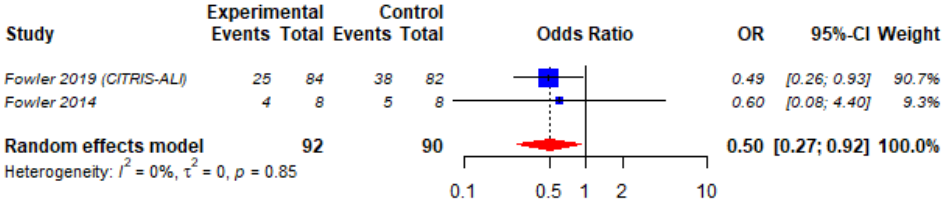
HD-vitC + GC + vitB1 vs GC



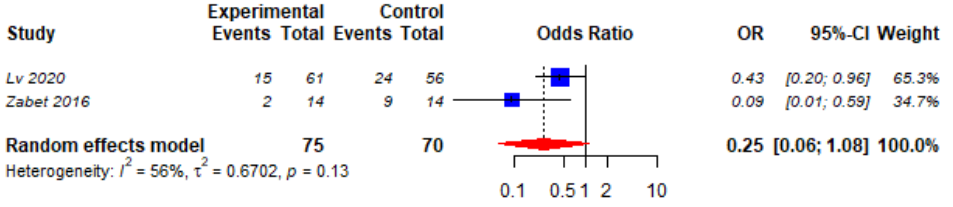
HD-vitC + GC + vitB1 vs placebo or usual care



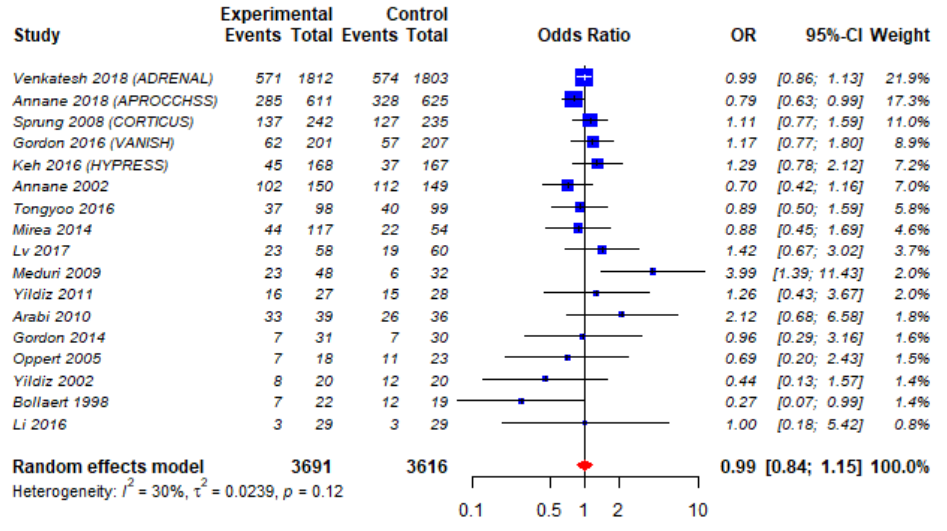
VHD-vitC vs placebo or usual care



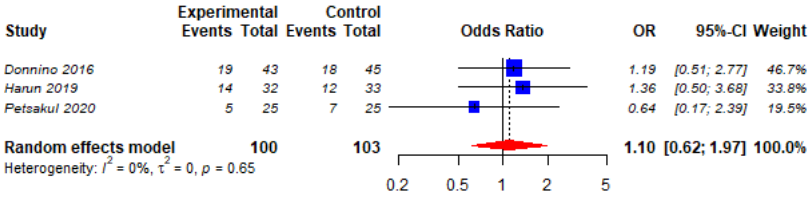
HD-vitC vs placebo or usual care



GC vs placebo or usual care



vitB1 vs placebo or usual care



12.3. Inconsistency

12.3.1. side-split

Number of loops = 2. GC denotes glucocorticoid; HD-vitC, high dose vitamin C; vitB1, vitamin B1: VHD-vitC, very high dose vitamin C; vitC, vitamin C; OR, odds ratio.

comparison	Number of studies	OR from NMA	OR from direct comparison	OR from indirect comparison	Ratio of ratios	p-value
HD-vitC+GC+vitB1 vs HD-vitC+vitB1	0	0.85	.	0.85	.	.
HD-vitC+GC+vitB1 vs placebo or usual care	6	1.05	1.07	0.98	1.1	0.7494
HD-vitC+GC+vitB1 vs VHD-vitC	0	2.1	.	2.1	.	.
HD-vitC+GC+vitB1 vs vitB1	0	0.95	.	0.95	.	.
HD-vitC+GC+vitB1 vs vitC	0	3.21	.	3.21	.	.
HD-vitC+vitB1 vs placebo or usual care	1	1.24	1.24	.	.	.
HD-vitC+vitB1 vs VHD-vitC	0	2.48	.	2.48	.	.
HD-vitC+vitB1 vs vitB1	0	1.12	.	1.12	.	.
HD-vitC+vitB1 vs vitC	0	3.78	.	3.78	.	.
VHD-vitC vs vitB1	0	0.45	.	0.45	.	.
VHD-vitC vs vitC	1	1.53	1.67	1.1	1.52	0.8505
VHD-vitC vs placebo/usual	2	0.5	0.5	.	.	.
HD-vitC vs HD-vitC+GC+vitB1	0	0.33	.	0.33	.	.
HD-vitC vs HD-vitC+vitB1	0	0.28	.	0.28	.	.
HD-vitC vs VHD-vitC	0	0.69	.	0.69	.	.
HD-vitC vs vitB1	0	0.31	.	0.31	.	.
HD-vitC vs vitC	0	1.05	.	1.05	.	.
HD-vitC vs placebo or usual care	2	0.34	0.34	.	.	.
vitC vs placebo or usual care	1	0.33	0.36	0.24	1.50	0.8505
GC vs HD-vitC	0	2.83	.	2.83	.	.
GC vs HD-vitC+GC+vitB1	2	0.93	0.99	0.91	1.1	0.7494
GC vs HD-vitC+vitB1	0	0.78	.	0.78	.	.
GC vs placebo or usual care	11	0.97	0.97	1.06	0.91	0.7494
GC vs VHD-vitC	0	1.94	.	1.94	.	.
GC vs vitB1	0	0.88	.	0.88	.	.
GC vs vitC	0	2.97	.	2.97	.	.
vitB1 vs vitC	0	3.37	.	3.37	.	.
vitB1 vs placebo or usual care	3	1.1	1.1	.	.	.

12.3.2. global

Q statistic to assess consistency under the assumption of a full design-by-treatment interaction random effects model = 0.14 (df=2, p=0.9338)

13. References

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